

Introduction & Hook

Ask students:

“When you touch a virtual object in VR and feel vibration in the controller, why does that happen?”

Explain:

In VR systems, interaction is not one-way.

When users perform an action, the system responds with **feedback** so that the user feels connected to the virtual environment.

This is called **user interaction and feedback**.

PART 1: User Interaction in VR

What is User Interaction?

Definition:

User interaction refers to the communication between the user and the virtual environment through input and feedback mechanisms.

Interaction occurs when the user:

- Moves their head
- Uses hand controllers
- Performs gestures
- Speaks commands
- Touches virtual objects

The VR system processes these inputs and generates feedback.

PART 2: Types of Feedback in VR

Feedback helps the user **understand the result of their action**.

There are three main types.

Visual Feedback

Visual Feedback

Definition:

Visual feedback is the information presented to the user through visual changes in the virtual environment.

Examples:

- Object changes color when selected
- Button highlights when pointed at
- Door opens after interaction
- Warning lights appear in simulations

Visual feedback helps users understand:

- Object state
- Interaction results
- Navigation cues

Example:

In VR training, pressing a switch causes a **machine animation** to start.

Audio Feedback

Audio Feedback

Definition:

Audio feedback provides sound cues to inform the user about interactions or environmental events.

Types of audio in VR:

- Spatial audio (3D sound)
- Interaction sounds
- Environmental sounds

Examples:

- Footsteps while walking
- Button click sound
- Machine noise
- Explosion in VR games

Benefits:

- Improves immersion
- Helps identify object location
- Supports user awareness

Example:

In VR training, when a machine starts, users hear the **motor sound**.

Haptic Feedback

Haptic Feedback

Definition:

Haptic feedback provides physical sensations such as vibration or force to simulate touch.

It is delivered through:

- VR controllers
- Haptic gloves

- Force feedback devices

Examples:

- Controller vibrates when grabbing object
- Resistance when pushing virtual button
- Feeling impact in VR games

Benefits:

- Enhances realism
- Improves interaction accuracy
- Increases user immersion

Example:

In VR surgery simulation, touching tissue produces **tactile resistance**.

PART 3: Interaction Loop

What is an Interaction Loop?

Definition:

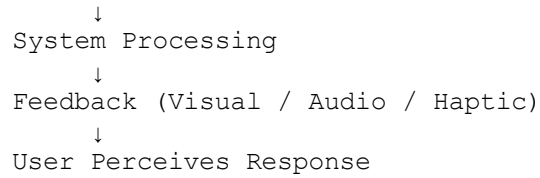
The interaction loop describes the continuous cycle between user input, system processing, and feedback.

Basic steps:

1. User performs an action
 2. System detects the input
 3. VR engine processes the event
 4. System provides feedback
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Interaction Loop Diagram

User Action
↓
Input Detection



This loop happens **continuously in real time**.

Example Interaction Loop

User grabs a virtual object:

1. User presses controller trigger
2. System detects trigger event
3. Object attaches to virtual hand
4. Controller vibrates
5. User sees object moving with hand

The cycle repeats every frame.

Real VR Application Example

VR training simulation:

- User presses emergency switch
- Machine stops
- Alarm sound plays
- Controller vibrates

This combines:

- Visual feedback
 - Audio feedback
 - Haptic feedback
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Recap & Questions

Key Points

- User interaction connects the user with the VR environment
 - Feedback helps users understand system responses
 - Three types of feedback: visual, audio, and haptic
 - Interaction loops continuously process user actions and responses
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Quick Questions

1. What is user interaction in VR?
2. Name three types of feedback used in VR.
3. What is haptic feedback?
4. Explain the interaction loop in VR systems.